

Knowledge Representation and Reasoning (AI3001)

Date: September 19, 2026

Course Instructor(s)

Muhammad Rafi

MidTerm - 1

Total Time (Hrs): 1

Total Points: 40

Total Questions: 4

Roll No

Section

Student Signature

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Attempt all the questions.

CLO 1: Understand the fundamentals of knowledge representation and reasoning in deterministic situations

Q1: Answer the following questions:

[Points: 2.5 X 4 =10]

- a. Define what is meant by Syntax and Semantic in the context of FOL? Give an example.

Syntax - it defines the formal rules for constructing well-formed formulas (WFFs) using symbols (constants, variables, functions, predicates, logical connectives, and quantifiers) without regard to their meaning. e.g. $\forall x \text{ man}(x) \rightarrow \text{mortal}(x)$

Semantics - it defines the meaning or interpretation of those formulas by mapping the syntactic symbols to elements, relations, and truth values (True/False) within a specific domain or model world. "Every man is mortal."

- b. Give two differences between Direct Resolution and Resolution by Refutation.

Resolution (Forward Proof) - Starts with the initial Knowledge Base (KB) and repeatedly applies resolution rules to generate new clauses until the target goal clause Q is explicitly derived. Terminates successfully when the exact goal literal/clause Q is derived $\text{KB} \vdash Q$.

Resolution by Refutation (Proof by Contradiction) - Negates the goal sentence ($\sim Q$), adds it to the KB, and resolves clauses searching for an explicit contradiction. Terminates successfully when the empty clause $\{\}$ is derived, signifying a contradiction $\text{KB} \cup \{\sim Q\} \vdash \{\}$.

- c. What is Strong AI? Give an example.

Strong AI (also known as Artificial General Intelligence / AGI) refers to a theoretical form of artificial intelligence that possesses human-level (or superior) cognitive abilities, self-awareness, and consciousness. Unlike Weak AI (which is specialized to perform specific tasks like playing chess, translating text, or recognizing images), a Strong AI system can - (1) Reason, solve problems, and think abstractly across any intellectual domain without prior task-specific programming, (2) Learn and apply knowledge autonomously in unfamiliar situations just as a

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human mind would and (3) Possess genuine understanding, consciousness, and subjective mental states (sentience).

d. Differentiate between Derivation and Entailment.

Feature	Entailment ($\models$)	Derivation ($\vdash$)
Fundamental Nature	A semantic relationship based on truth, models, and meaning in possible worlds.	A syntactic relationship based on formal manipulation of symbols and inference rules.
Core Question	"In every world/model where the knowledge base is true, is the sentence also true?"	"Can the sentence be generated from the knowledge base using a sequence of proof steps?"
Determining Factor	Depends on truth values across interpretations, independent of any specific algorithm.	Depends on algorithmic steps executed by an inference engine or theorem prover.

CLO 2: Understand the challenges in representing knowledge and reasoning under uncertainty

Q2: Answer the following questions.

[Points: 5+5 =10]

- a. Why is Skolemization sound for refutation, but not truth-preserving (equisatisfiable vs. logically equivalent)?

Skolemization replaces existentially quantified variables $\exists$ with fresh Skolem constants or functions. Not Logically Equivalent (Not Truth-Preserving) - Two formulas are logically equivalent $\phi \equiv \psi$ if they have the exact same truth value under every possible interpretation (model). Consider the simple sentence: $\exists x \mid \text{Dog}(x)$ If we Skolemize this by replacing x with a new constant C , we get: $\text{Dog}(C)$ - $\text{Dog}(C)$ asserts that a specific, named entity C is a dog. Whereas, $\exists x \text{ Dog}(x)$ asserts only that at least one dog exists somewhere in the domain, without specifying who it is. Hence, Skolemization is not truth-preserving and does not preserve logical equivalence.

Equisatisfiable (Sound for Refutation) - Two formulas ϕ and ψ are **equisatisfiable** if ϕ has a model if and only if ψ has a model. If $\exists x, P(x)$ is satisfiable $\iff$ If $P(C)$ is satisfiable. Because Skolemization guarantees equisatisfiability: Original Formula Set is Unsatisfiable iff Skolemized Clause Set is Unsatisfiable.

- b. For each of the following pairs of atomic expressions, determine if they are unifiable. If they are unifiable, provide the Most General Unifier (MGU). If they are not unifiable, explain precisely why (e.g., constant mismatch, occurs check failure, function arity):

- i. $R(x, F(x))$ and $R(y, y)$

Arg 1: Unify variable x with variable $y \Rightarrow \theta_1 = \{x / y\}$.

Arg 2: Apply θ_1 to $F(x)$, yielding $F(y)$. Attempt to unify variable y with $F(y)$.

Performing the Occurs Check – variable y occurs inside the term $F(y)$, unifying them would result in an infinite cyclic structure for $y = F(F(F(\dots)))$.

Result – Not unifiable

- ii. $S(x, A, F(G(y)))$ and $S(B, y, F(z))$

Arg 1: Unify variable x with constant $B \Rightarrow \theta_1 = \{x / B\}$.

Arg 2: Unify constant A with variable $y \Rightarrow \theta_2 = \{x / B, y / A\}$.

Arg 3: Apply θ_2 to $F(G(y))$, yielding $F(G(A))$. Unify $F(G(A))$ with $F(z)$, Unify $G(A)$ with variable $z \Rightarrow z / G(A)$ (Occurs check passes as z does not appear in $G(A)$).

Hence Most General Unifier (MGU) is $\theta = \{x/B, y/A, z/G(A)\}$

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CLO 4: Development of hybrid approaches by synergizing the existing framework to solve complex decision-making problems

Q3: Answer the following questions briefly to the point.

[Points: 5+5 =10]

Every citizen who is a senior citizen is exempt from local property tax. If a citizen is an ex-army, they are also classified as senior citizen. John is a citizen and served in army.

- a. Convert it into a FOL statements.

SeniorCitizen(x) - x is a senior citizen.
TaxExempt(x) - x is exempt from local property tax.
ExArmy(x) - x is an ex-army member.
From sentence 1 - $\forall x (\text{SeniorCitizen}(x) \Rightarrow \text{TaxExempt}(x))$
From sentence 2 - $\forall x (\text{ExArmy}(x) \Rightarrow \text{SeniorCitizen}(x))$
From sentence 3 - $\text{ExArmy}(\text{john})$.

- b. Convert each FOL from part(a) into CNF.

FOL1 - $\sim \text{SeniorCitizen}(x_1) \vee \text{TaxExempt}(x_1)$
FOL2 - $\sim \text{ExArmy}(x_2) \vee \text{SeniorCitizen}(x_2)$
FOL3 - $\text{ExArmy}(\text{john})$.

- c. Illustrate the KB with the knowledge statements in FOL(CNF)

KB
C1: $\sim \text{SeniorCitizen}(x_1) \vee \text{TaxExempt}(x_1)$
C2: $\sim \text{ExArmy}(x_2) \vee \text{SeniorCitizen}(x_2)$
C3: $\text{ExArmy}(\text{john})$.

- d. Use Resolution to prove "John is Exempted from local property tax. e.g. $\text{TaxExempt}(\text{john}) = \text{true}$.

Proof by Resolution

Goal to prove: C4: $\text{TaxExempt}(\text{John})$, Negated Goal C 4: $\sim \text{TaxExempt}(\text{John})$

Step 1: Resolve Negated Goal (C4) with C1 for $\theta = \{x_1 / \text{John}\}$, we get
C5: $\sim \text{SeniorCitizen}(\text{John})$

Step 2: Resolve C5 with C2 for $\theta = \{x_2 / \text{John}\}$, we get C6: $\sim \text{ExArmy}(\text{John})$

Step 3: Resolve C6 with C3 for $\theta = \{\}$, we get C7: $\{\}$ (Empty Clause / Contradiction)

Hence proved C4: $\text{TaxExempt}(\text{John})$ is consistent with KB.

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CLO 2: Understand the challenges in representing knowledge and reasoning under uncertainty

Q4: Consider the following knowledge base in Prolog, write the relevant rules and

```
% Facts: Observed symptoms for patients
symptom(alice, fever).
symptom(alice, cough).
symptom(alice, loss_of_smell).

symptom(bob, fever).
symptom(bob, sore_throat).
symptom(bob, runny_nose).

symptom(charlie, runny_nose).
symptom(charlie, sneezing).
symptom(charlie, cough).
```

- a. A patient is suspected of having a seasonal_allergy if they have a runny nose and sneezing, but do not have a fever. [2.5]

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suspect_disease(Patient, seasonal_allergy) :-
    symptom(Patient, runny_nose),
    symptom(Patient, sneezing),
    \+ symptom(Patient, fever).
```

- b. A patient is suspected COVID-19, if the patient has fever, cough, and loss of smell. [2.5]

```
suspect_disease(Patient, covid19) :-
    symptom(Patient, fever),
    symptom(Patient, cough),
    symptom(Patient, loss_of_smell).
```

- c. Prove by resolution that patient “charlie” has seasonal_allergy from the KB. Show all steps on resolution using unification and skolemization. [5]

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Step 1 - Call: suspect_disease(charlie, seasonal_allergy)
Step 2 - Call:symptom(charlie,runny_nose)
Exit:symptom(charlie,runny_nose)
Step 3 -Call:symptom(charlie,sneezing)
Exit:symptom(charlie,sneezing)
Step 4 - Call:symptom(charlie,fever)
Fail:symptom(charlie,fever)
Step 5 -Redo:suspect_disease(charlie,seasonal_allergy)
Exit:suspect_disease(charlie,seasonal_allergy)
true
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<The End. >